

A263 – Dirac Equation in FFGFT: Clifford Algebra, Spin, and Mass Elimination

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.1 Purpose

[STIPULATION] This document develops the Dirac equation in FFGFT completely: the Clifford algebra as fundamental structure, the transition from the standard Dirac to the T0 formulation, mass elimination (physics formulable without mass parameters), spin as topological property, and the modified Dirac equation from the time-field Lagrangian (A142). The document supplements A075 (field theory, Schrödinger limit) with the complete relativistic structure.

.2 The Clifford Algebra as Fundamental Basis

[B] The standard Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ uses 4×4 γ -matrices as a representation. In FFGFT the fundamental structure is the Clifford algebra:

$$\mathbf{e}_\mu\mathbf{e}_\nu + \mathbf{e}_\nu\mathbf{e}_\mu = 2g_{\mu\nu}.$$

The γ -matrices are one representation – they are not eliminated but traced back to their geometric origin. That is the difference: in the SM γ -matrices are the foundation; in FFGFT it is the Clifford algebra, with γ -matrices being only one representation.

[B] What changes in T0? The standard Dirac equation has m as a fixed parameter. In T0 the mass $m(x, t)$ is a dynamic field:

$$[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x, t)]\psi = 0,$$

with time-field connection $\Gamma_\mu^{(T)} = -\partial_\mu m/m$ (A142). In the limit $\partial_\mu m \rightarrow 0$ (constant mass) this reduces to the standard Dirac equation. (Corrected on 30 Sept. 2026: previously $\Gamma_\mu^{(T)} = \partial_\mu m/m^2$ – with $\mathcal{T} = 1/m$ one has $\partial_\mu \mathcal{T} = -\partial_\mu m/m^2$, hence $\Gamma_\mu^{(T)} = \mathcal{T}^{-1}\partial_\mu \mathcal{T} = -\partial_\mu m/m$.)

.3 Spin as Topological Property

[B] In FFGFT spin is not an intrinsic property but a topological one: spin- $\frac{1}{2}$ corresponds to the torus winding $(n_\theta, n_\phi) = (1, 1)$. The 720° symmetry (two rotations return a spinor to its initial state) follows directly from the topology of the torus – it need not be postulated.

[B] A relativistic spinor state in the torus representation:

$$\Phi(x, t) = \rho(x, t) e^{i\theta(x, t)} \cdot \chi(\sigma),$$

with spin component $\chi(\sigma)$ from the geometric knot modes (A075). The quantum number $j = l \pm \frac{1}{2}$ for fermions follows from the winding structure.

.4 Mass Elimination: Physics Without Mass Parameters

[K] In the T0 framework all observables can be formulated as ratios that require no absolute mass parameter. The basic idea: instead of carrying m in SI units as a parameter, everything is expressed through ξ and time intervals.

[K] Systematic mass elimination. From the T0 duality $\tilde{T} \cdot m = 1$ it follows that $m = 1/\tilde{T}$. Substituting into the Dirac equation:

$$[i\gamma^\mu \partial_\mu - 1/\tilde{T}(x, t)]\psi = 0.$$

The mass parameter disappears from the equation; what remains is the time field $\tilde{T}(x, t)$ as a dynamic quantity. This is the fully mass-free T0 formulation.

[S] Philosophical consequence. If all physics is formulable in ratios (without absolute masses), then masses are not fundamental objects but auxiliary constructs of measurement. What is fundamental: time intervals and the geometry of the torus. This is the same thesis as A085 (natural units) in relativistic language.

.5 Mass-Proportional Coupling

[K] In the T0 Lagrangian the time field couples mass-proportionally: $g_T^\ell = m_\ell \cdot \xi$. This is the mechanism behind the g -2 calculation (A138): the coupling is not a free Yukawa constant but fully determined by the mass and ξ . The one-loop formula for the anomalous magnetic moment follows directly from this mass-proportional coupling.

.6 Ratios vs. Absolute Values in the Dirac Structure

[B] The same distinction as in A261 and A138: ratios of Dirac corrections (e.g. a_μ/a_e) are correction-free and follow from the structure of the Clifford algebra and the quantum numbers. Absolute values of the g -2 terms require the SI bridge constants and K_{frac} (A138).

.7 Verification Script

- Numerical verification: standard Dirac equation as limiting case of the T0 form ($\partial_\mu m \rightarrow 0$); spin- $\frac{1}{2}$ from winding (1,1); mass-free formulation for simple cases: `python/A_Serie_Skripte/a263_dirac.py`

.8 Open Questions

The complete mass-free formulation for all interactions is missing. Mass elimination is shown for free fields; for interaction terms it has not been systematically carried out.

The quark sector is missing. The Dirac equation for quarks with dynamic mass and colour charge is not developed at the same level as for leptons (A150).

The relation to QED renormalisation is not worked out. How the T0 Dirac equation relates to perturbative QED (Feynman rules, loop integrals) is not systematically developed (A075, open Hopf algebra).

.9 Supersedes

This document subsumes: Doc. 051 (T0 Dirac equation, Clifford algebra, spin as topology, mass-proportional coupling, ratios vs. absolute values), Doc. 052 (systematic mass elimination, complete mass-free T0 formulation), Doc. 053 (mass elimination in the Dirac Lagrangian, ratio-based physics), Doc. 050 (simplified Dirac equation).

.10 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.